

AI Network Troubleshooter Report

Filename: advanced_capture1.pcap

Packet Count: 2201

Protocols

ARP: 4

TCP: 2157

DNS: 12

MDNS: 4

NTP: 4

ICMP: 20

AI Diagnostic Report

1. Root cause summary:

The packet capture reveals a high volume of TCP traffic with an unusual pattern: out of 1023 SYN packets sent, only 6 SYN-ACK packets were received back. This indicates a very high number (1017) of possible failed TCP connection attempts, suggesting that the host at 172.17.208.219 is attempting to initiate TCP connections but not successfully completing handshakes. This likely points to network connectivity issues, firewall or ACL blocking, or a service on the destination not responding. The large number of failed connections is the root cause of the observed communication failure.

2. Security observations:

- The source IP 172.17.208.219 is generating a very large number of TCP SYN packets with minimal successful handshakes, which could indicate a SYN flood attack or scanning activity.
- External IP 45.33.32.156 is also a top talker with a similarly high packet count, which may be involved in malicious activity or heavy communication.
- DNS queries include reverse DNS lookups for IPs and a lookup to "scanme.nmap.org" — a domain often used for scanning and testing, which may imply internal scanning or reconnaissance.
- Overall, the traffic pattern shows potential reconnaissance or scanning behavior, which needs further analysis for possible intrusion attempts.
- The presence of multiple protocol types (ARP, ICMP, MDNS, NTP) is normal but should be monitored for abnormalities.

3. Troubleshooting recommendations:

- Verify firewall and ACL configurations to ensure legitimate TCP traffic is allowed and to identify if blocking causes failed connections.
- Investigate the services on the destination IP addresses for availability and responsiveness.
- Check for possible SYN flood attacks or port scanning on 172.17.208.219 by monitoring connection rates and patterns.
- Perform deeper packet inspection of the failed TCP connections to determine destination ports and attempt to identify the application or service involved.
- Review logs on the endpoints and network devices for any alerts related to connection failures or security events.
- Conduct vulnerability scans and network behavior analysis to validate if the traffic is malicious or a result of misconfigurations.
- Educate users or administrators about suspicious DNS queries including "scanme.nmap.org" to understand if security testing is authorized.

4. Severity assessment:

The severity is medium to high given the large number of failed TCP connection attempts which can impact network performance and indicate possible malicious scanning or denial of service attempts. Immediate investigation is warranted to rule out or mitigate security threats and resolve communication issues affecting service availability. If this is indeed an attack, the risk of network disruption and security breach increases significantly.